DOCKER & KUBERNETES

From Containers to Orchestration

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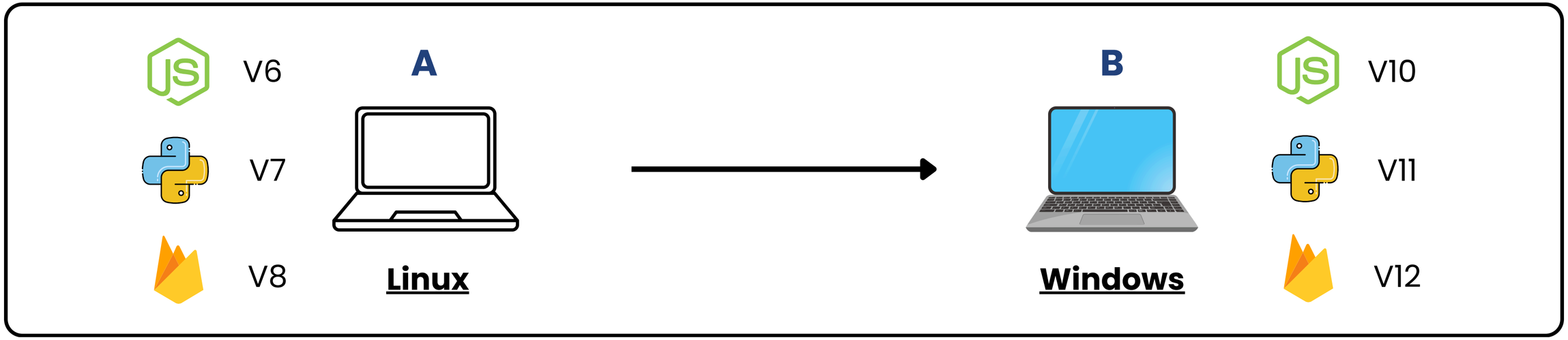
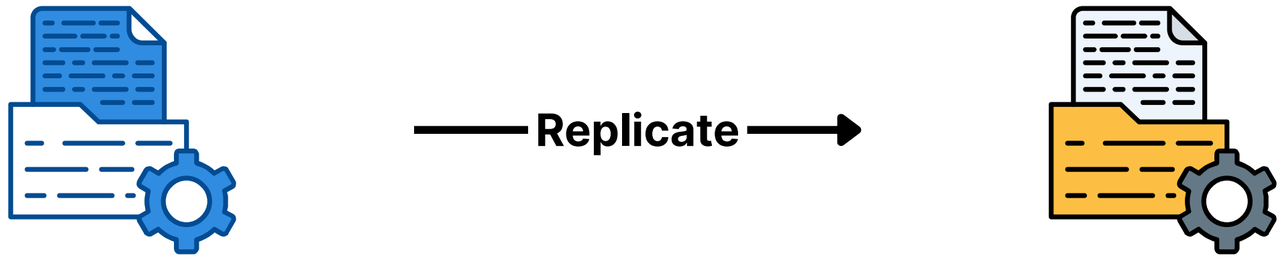
DOCKER?



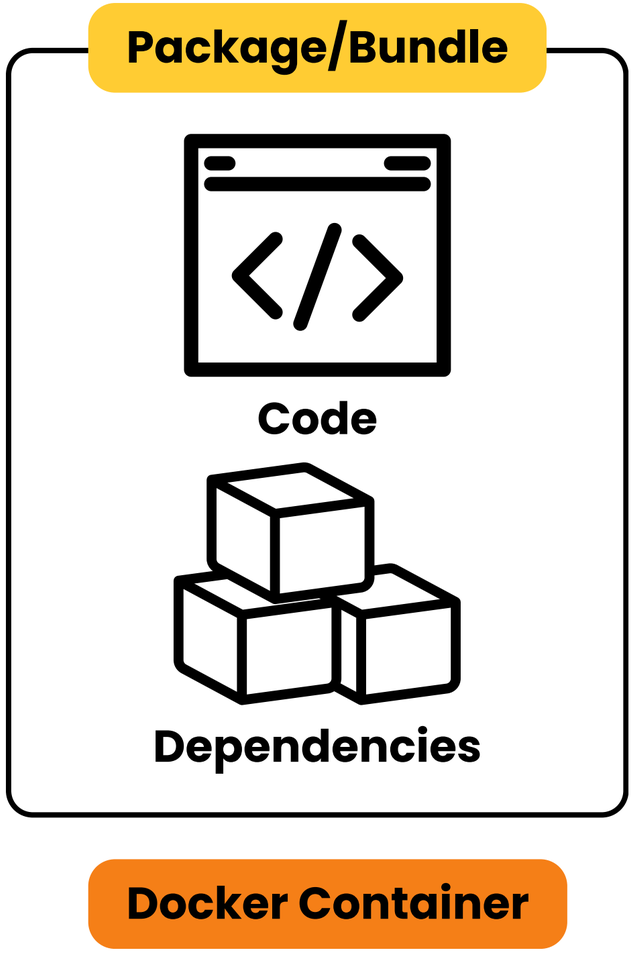
Where is the problem?



Environment of System 1

Environment of System 2

DOCKER



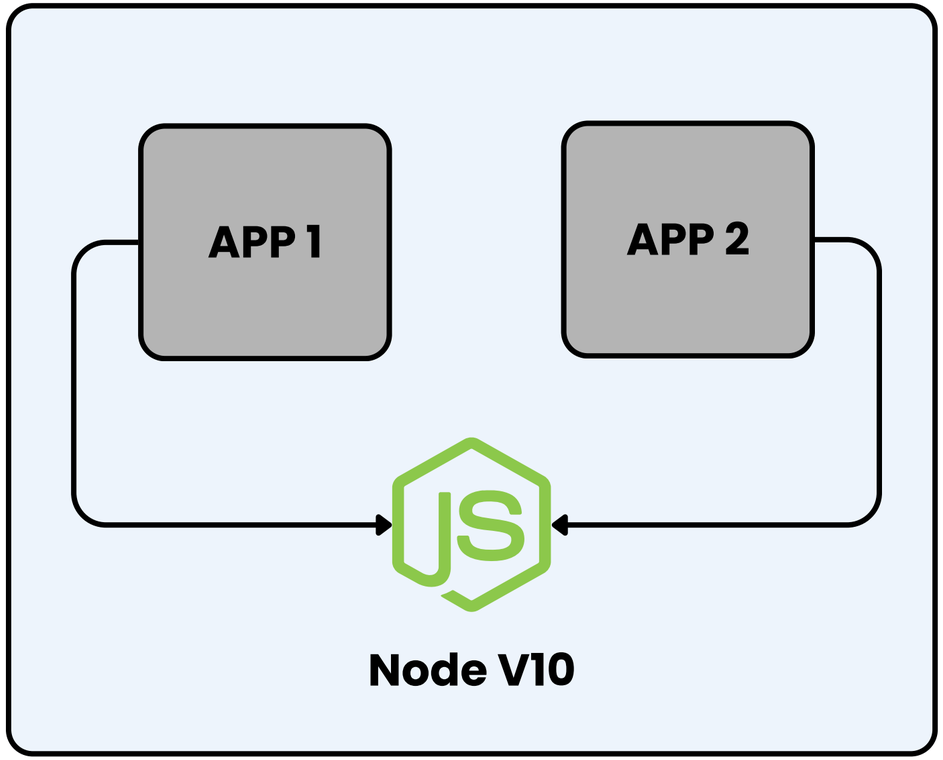
Docker is the service that build, destory, update and manage these containers

System Independent

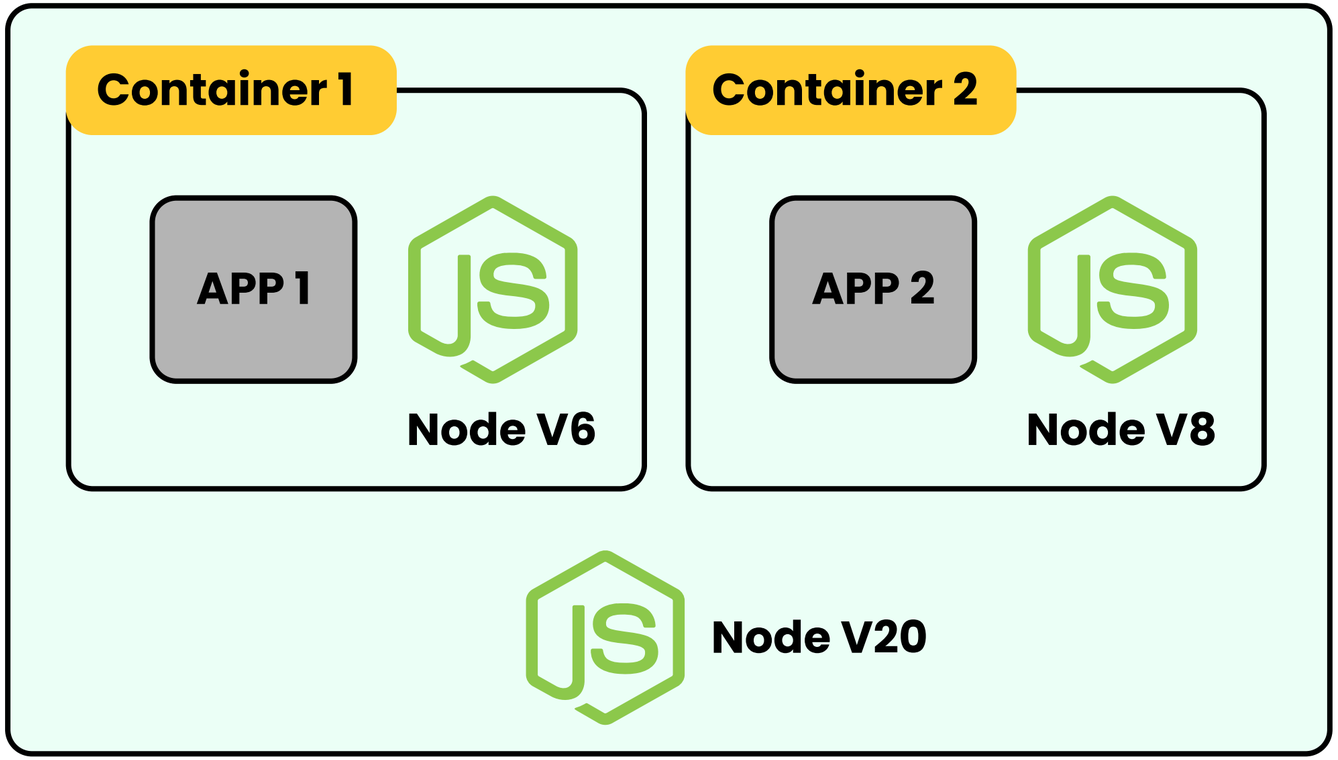
1 Single Unit

Portable & Lightweight

WITH AND WITHOUT DOCKER

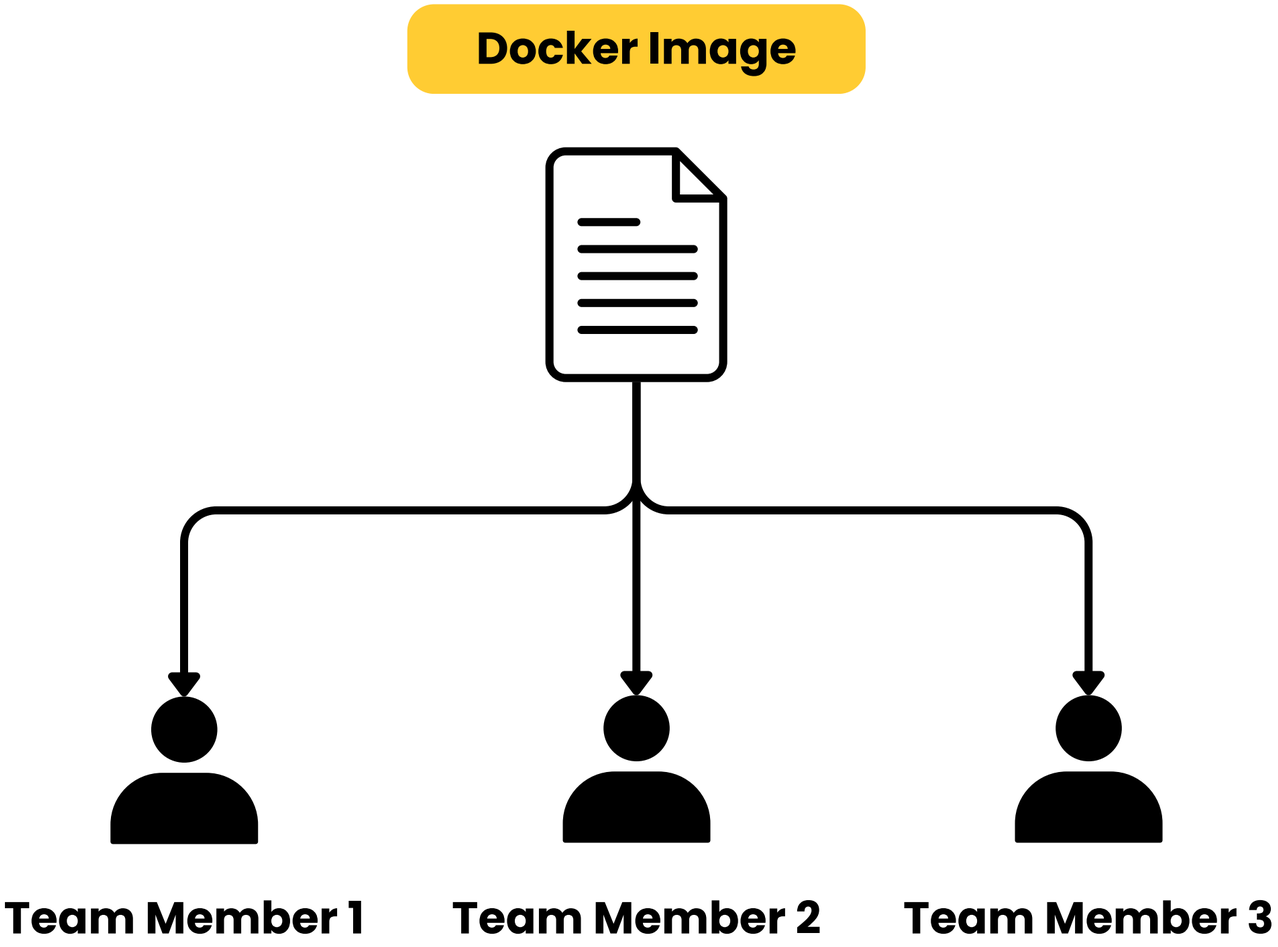


System without Docker



System with Docker

DOCKER IMAGE



Docker Image is just an executable file

Instructions about how the container is to be built

What is the code and what are the dependencies

What local environment should look like

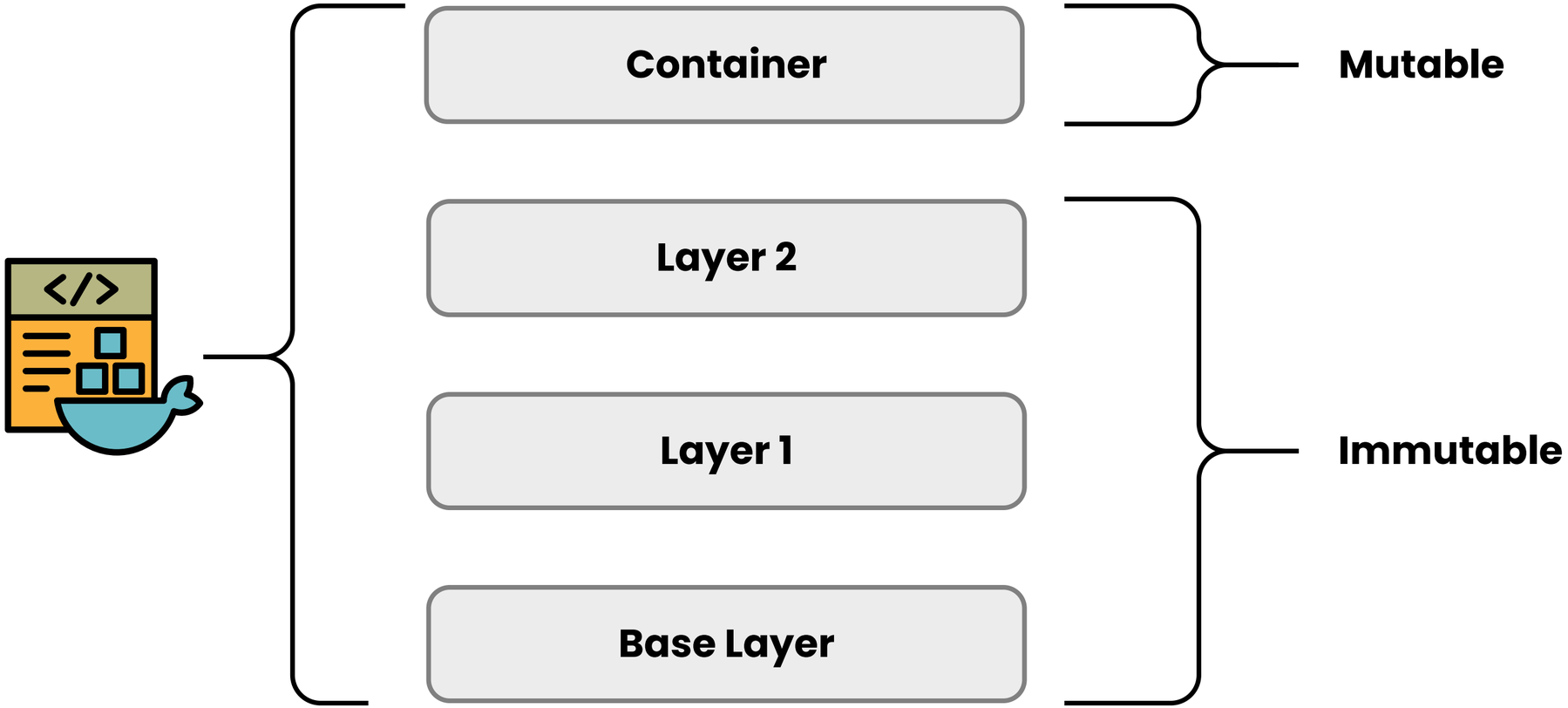
DOCKER DESKTOP AND DOCKER HUB

Docker Desktop provides an interface on our local system to manage containers and images

Docker Hub is used to store the built images and later pull them

LAYERS IN DOCKER

Docker Image is Collectio of Layers.



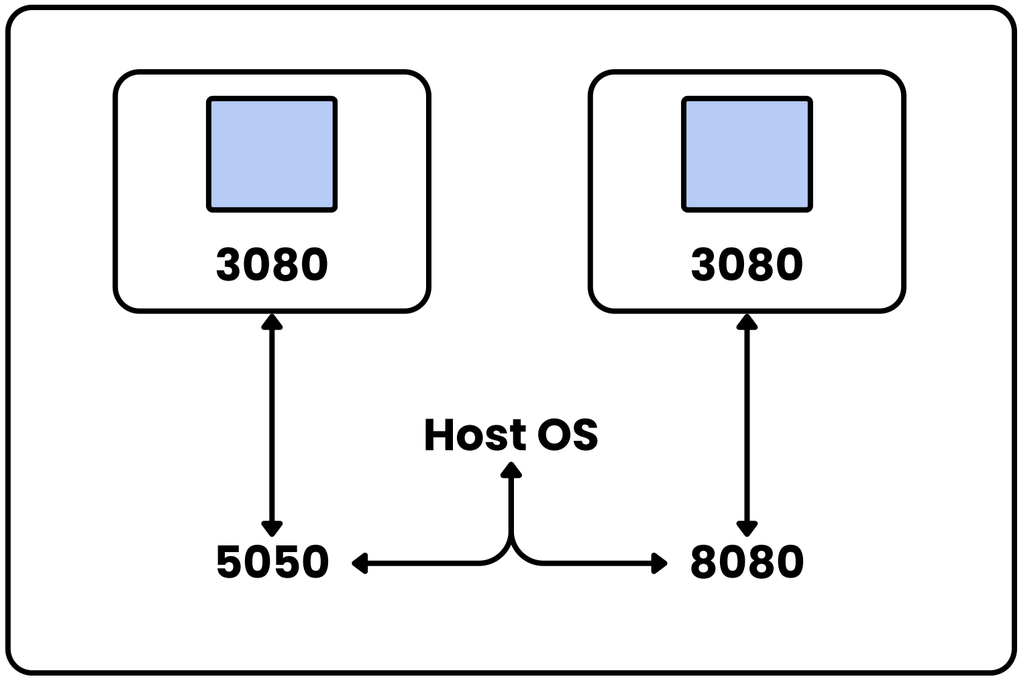
PORT BINDING IN DOCKER

Docker creates a network namespace for every container.

Each namespace has its own isolated network stack (ports, interfaces, IP)

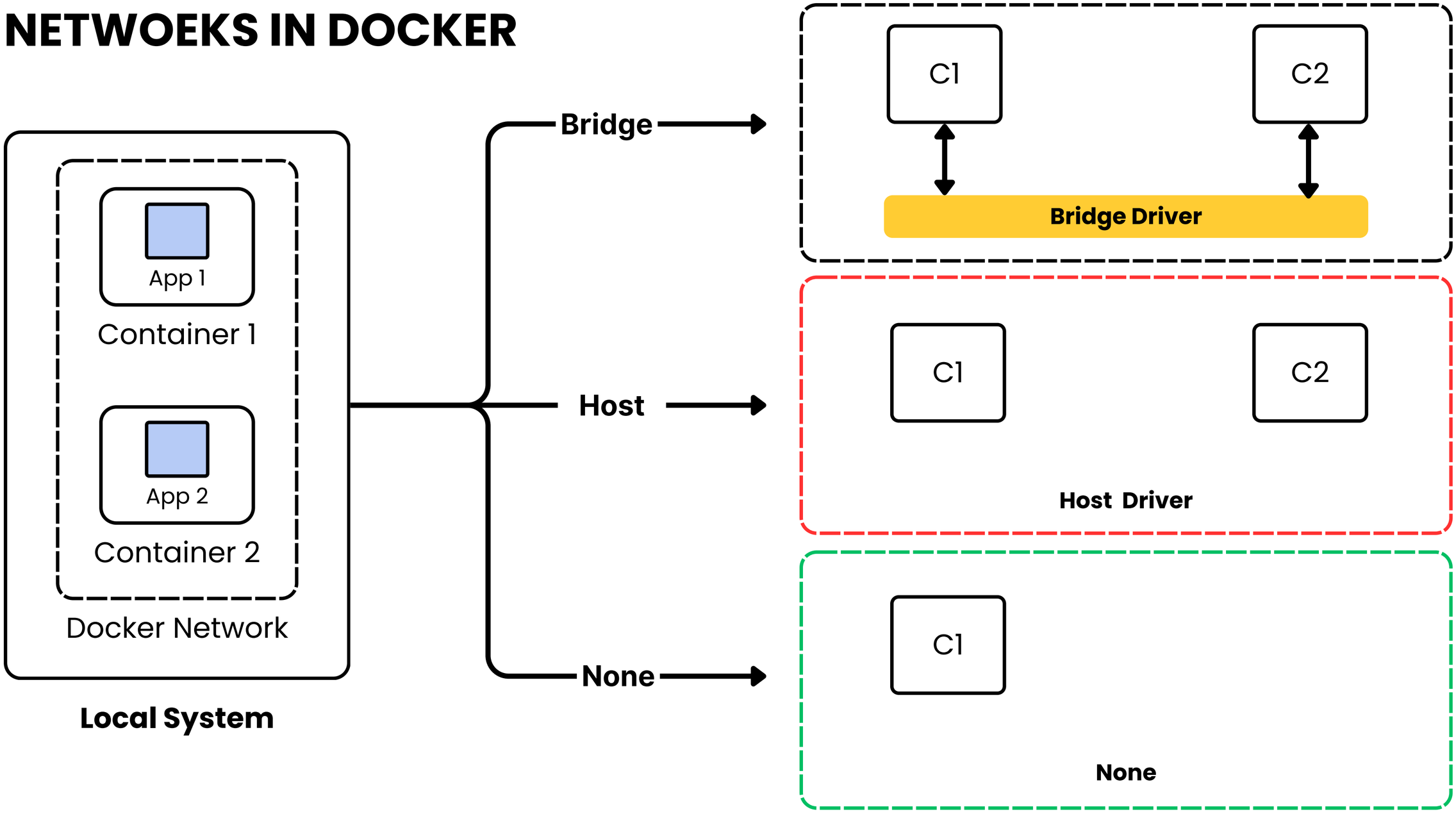
Containers uses the ports inside the Networkspace.

The host OS (via Docker) maps host ports to container ports, allowing external traffic to reach containers

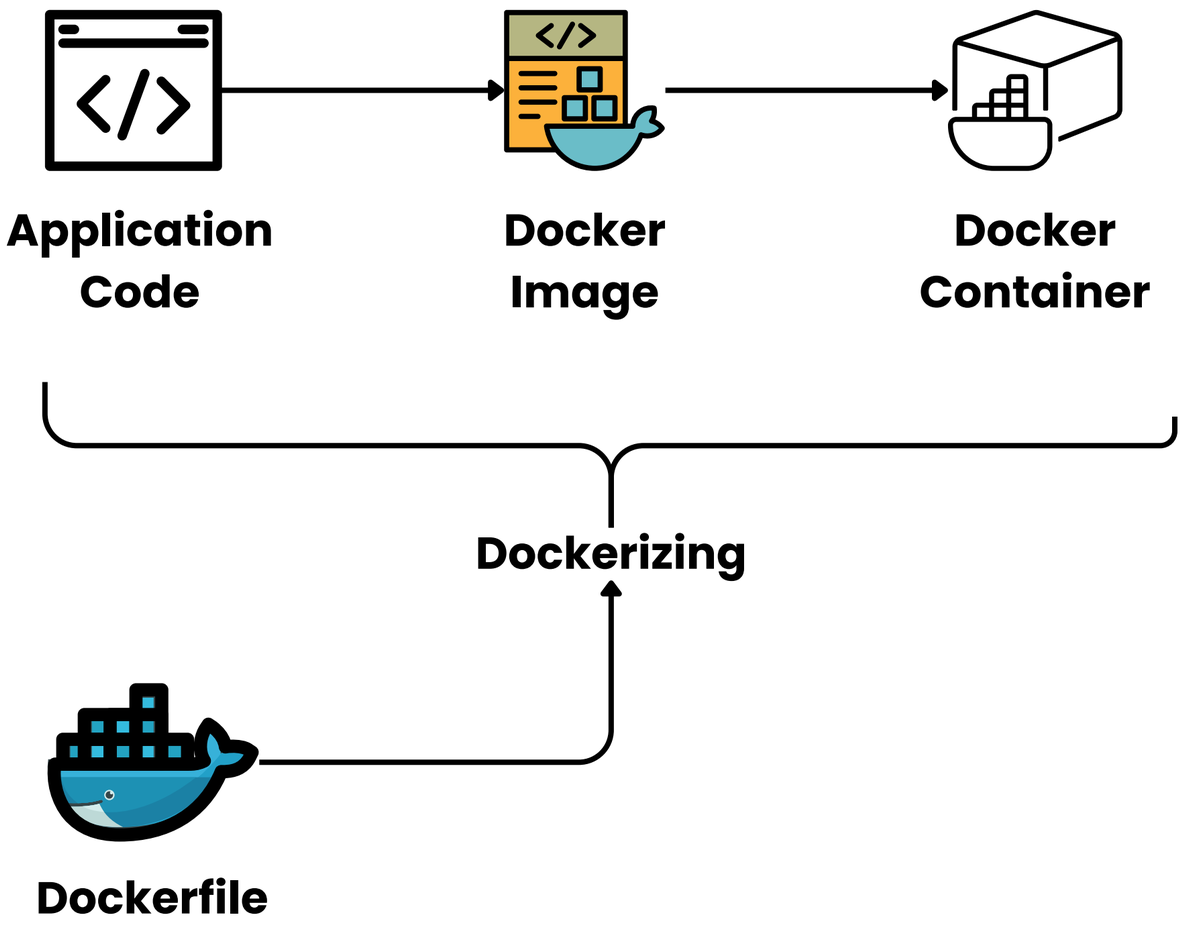


Local System

NETWOEKS IN DOCKER

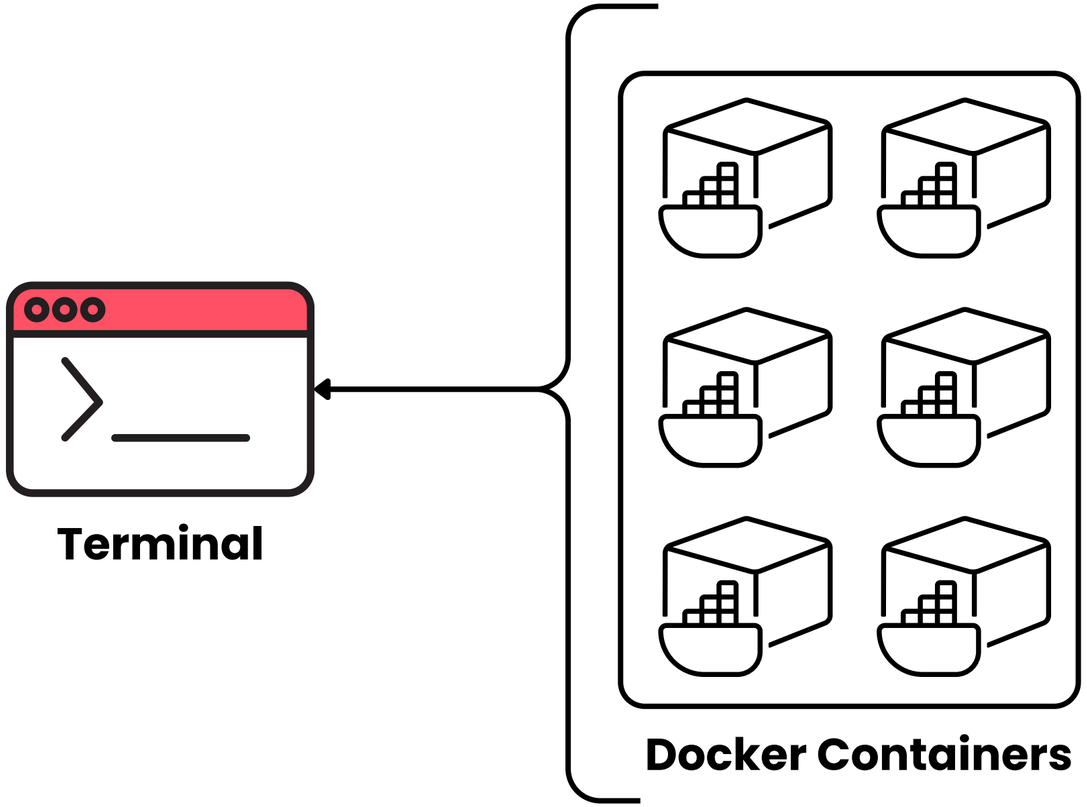


Dockerfile



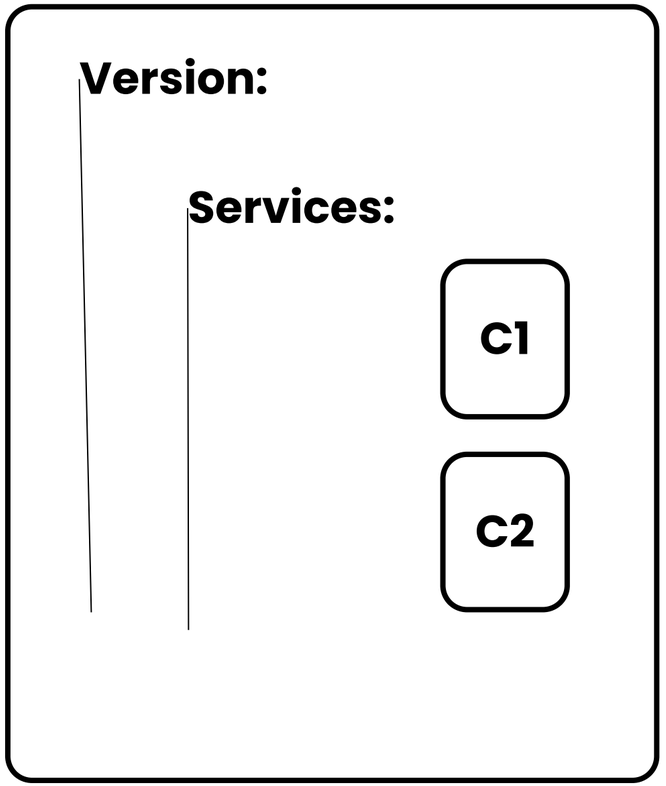
FROM	Name of base image (Node or Python)
WORKDIR	All commands will be executed in this dir. and files will be copied here
COPY	Copy the files and commands from your host to image
RUN	Commands that we want to execute. (Multiple)
CMD	Command to start the application
ENV	Set environment variables

Docker Compose

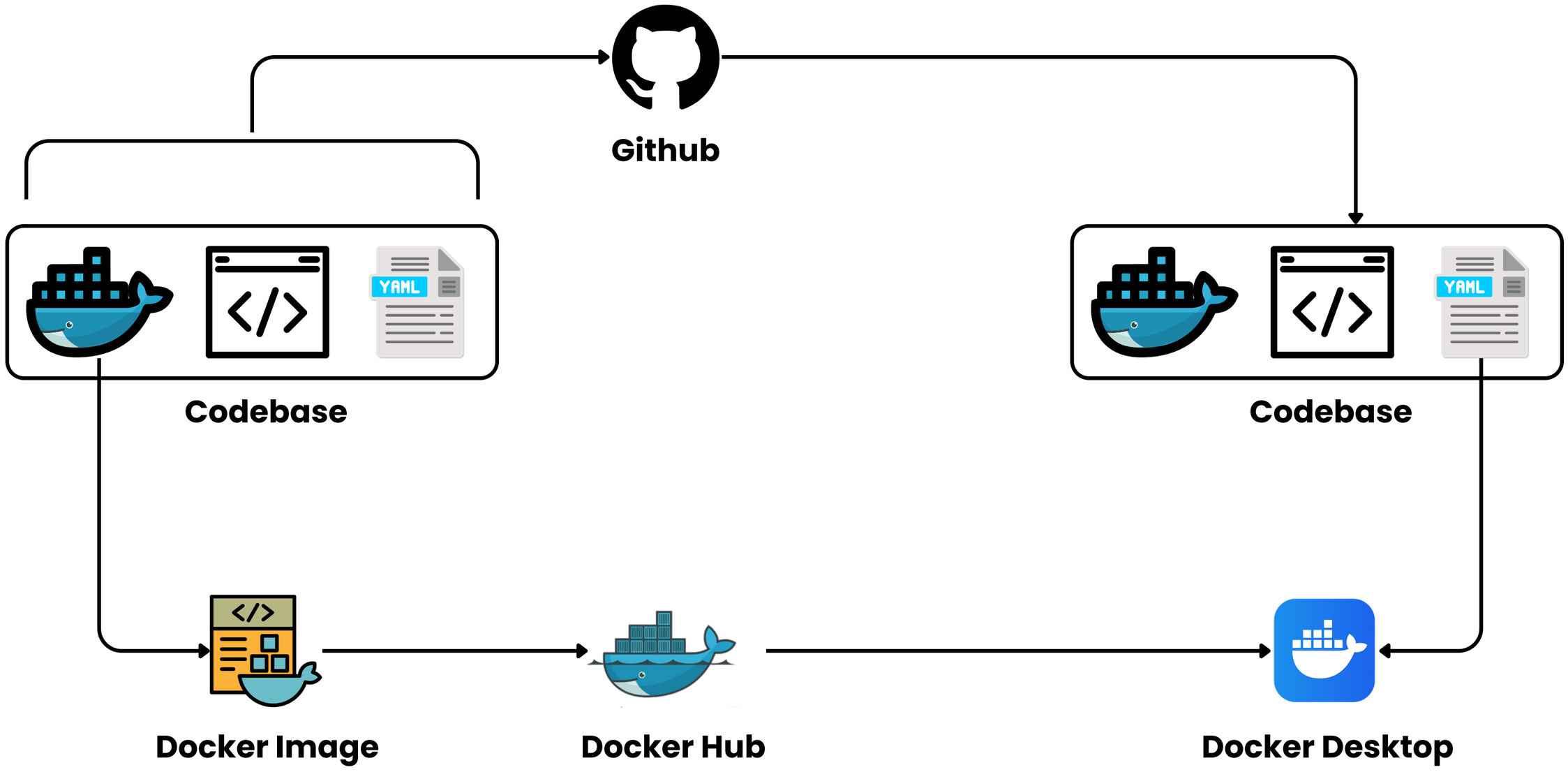


Complex

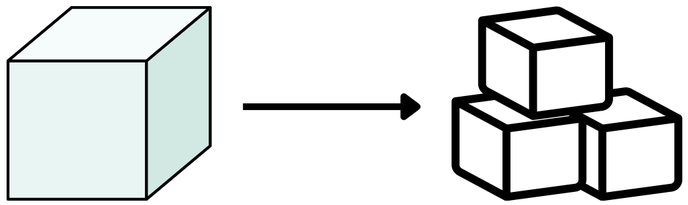
.YAML File



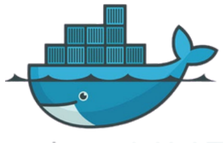
Development Flow



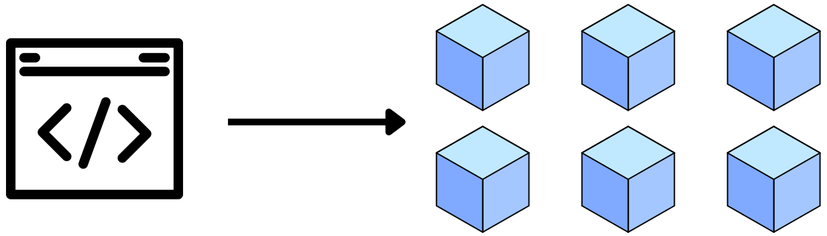
KUBERNETES



Monolithic **Microservices**

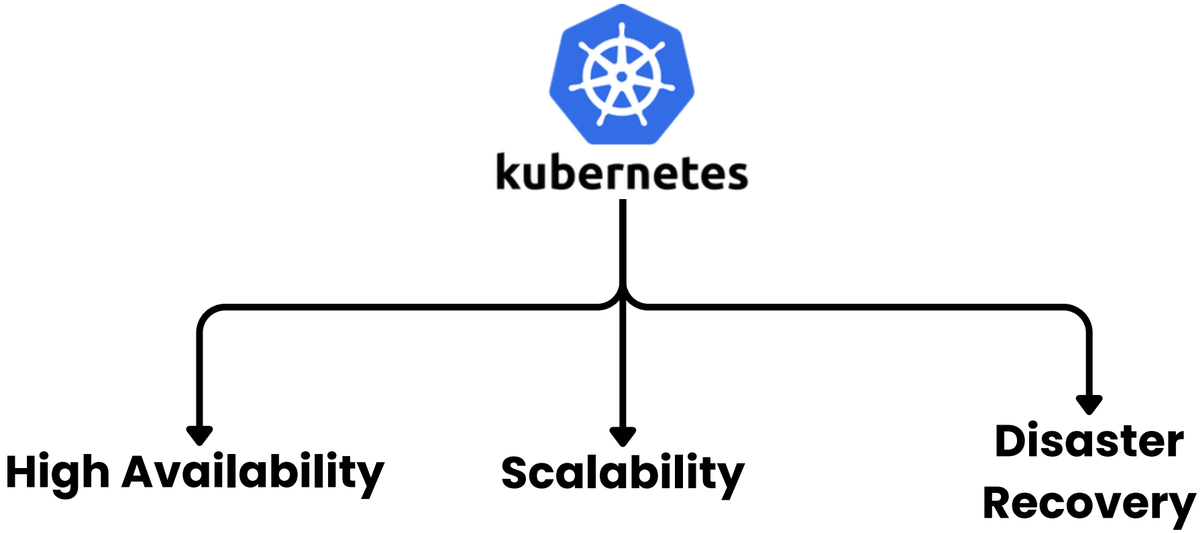


Containerization Microservices



How to manage 1000's of containers?

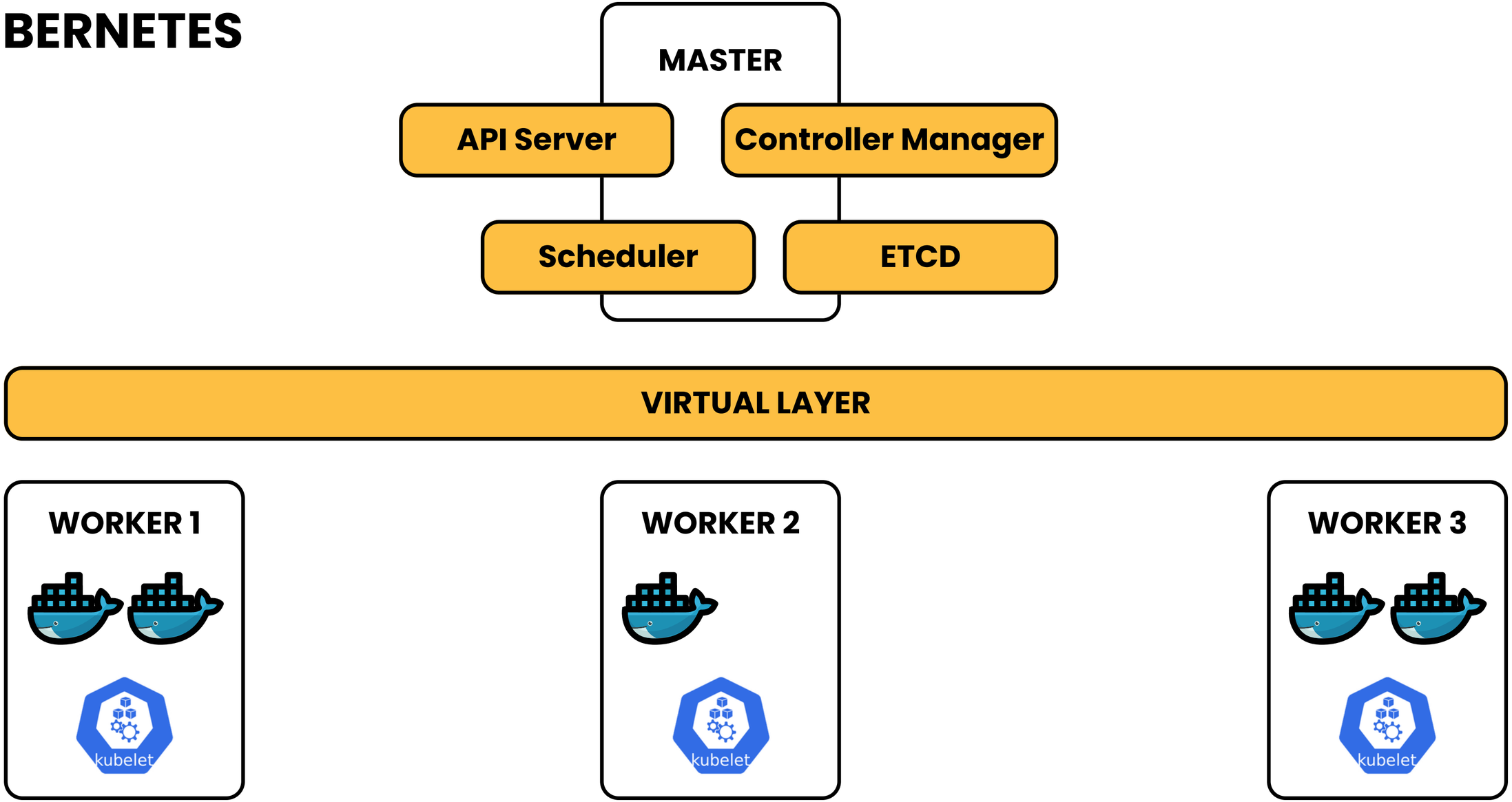
There is a need of orchestration tool?



Kubernetes works on a cluster

- A cluster = group of machines
- A cluster has: Control Plane (Master), Worker Nodes

KUBERNETES


MASTER
API Server
Controller Manager
Scheduler
ETCD
VIRTUAL LAYER
WORKER 1
kubelet
WORKER 2
kubelet
WORKER 3
kubelet

THANK YOU!

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